

Futures Pricing

Models And Comparisons

FINM 37000 - Futures And Related Derivatives

Eric Patterson, Fall 2025

Cost-of-carry Model

Financing a spot position

- Borrow to buy asset for S_0
- Holding until time T leads to having asset worth S_T and owing

$$V_{0,T} = S_0 + I_{0,T} + C_{0,T} - B_{0,T}$$

- $I_{0,T}$: interest paid on borrowing
- $C_{0,T}$: additional costs specific to the asset esp. storage
- $B_{0,T}$: benefits from the asset (e.g., dividends, interest payments)

Cost-of-carry Model

Offsetting with a forward

- Sell a forward at time 0 for price F_0 (not value) to offset the financed purchase of the asset
- At time $t = 0$, forward has value 0, asset $+S_0$, borrowing $-S_0$
- At time $t = T$,
 - Forward value $S_T - F_0$, short position is $F_0 - S_T$
 - Total position value $(F_0 - S_T) + S_T - V_{0,T}$
 - No arbitrage implies $F_0 - V_{0,T} \leq 0$, that is

$$F_0 \leq S_0 + I_{0,T} + C_{0,T} - B_{0,T}$$

Cost-of-carry Model

Bounding Forward Price From Below

- When can we reverse the cost-of-carry trade?
 - Need to be able to short the asset
 - Typically would be responsible for paying benefits like dividends
 - Typically do not need to offset the buyer's costs (e.g., they store it)

$$F_0 \geq S_0 + I_{0,T} - B_{0,T}$$

- If costs are negligible, $F_0 = V_{0,T}$ by considering both bounds.

Cost-of-carry Model

Convenience Yield

Difference between forward prices and the cost-of-carry is termed the (accrued) convenience yield $Y_{0,T} \geq 0$:

$$F_0 = S_0 + I_{0,T} + C_{0,T} - B_{0,T} - Y_{0,T}$$

Positive convenience yield reflects the desirability of holding the asset beyond the value already reflected in S_0 and $B_{0,T}$.

Extremely common to transform all of these to continuously compounded rates, with y the convenience yield, r the interest rate, u the rate of non-interest costs, q the rate of benefits, and c the cost of carry:

$$F_0 = S_0 e^{(r+u-q-y)T} = S_0 e^{(c-y)T}$$

WARNING 1: Sometimes $c - y$ is called the cost-of-carry, and other times just c , i.e., the costs that are not part of the convenience yield.

WARNING 2: Please check your sources as these letters are not standardized. Sometimes c is convenience yield, sometimes s is storage cost, etc.

It is also common to assume these rates are constant, but that is an additional assumption. There is really an implicit dependence on the period $[0, T]$ for all of these rates.

Cost-of-carry Model

Offsetting with a ~~forward~~-future

- Sell a future at time 0 for price F_0 , value 0, to offset the financed purchase of the asset
- At time $t = 0$, futures has value 0, asset $+S_0$, fund margin M_0 , borrow $-S_0 - M_0$
- At time $t = 1$ (assume units of days),
 - Daily mark-to-market margin: $M_1 = (F_1 - F_0) + M_0$
 - If $F_1 - F_0 \leq 0$, may need to add funds.
 - No-arbitrage arguments get messy.
 - Value of futures position resets to 0
 - Return on margin account $M_0 R_{M,0}$, where $R_{M,0}$ is the daily return on the margin account from day 0.

Basis Trade

Spot vs. Futures or Forwards

- Basis $S_t - F_t$ (or $F_t - S_t$ in some market conventions)
- Key to trading opportunities (the basis trade):
 - Understanding costs and benefits now and in the future
 - Data on storage costs, dividends, etc.
 - Understanding factors affecting convenience yield
 - Expectations about future supply and demand
 - Delivery optionality
- Access to execution across markets (e.g., physical markets, FX markets, bond markets, equity markets)

Basis Risk

Spot vs. Futures or Forwards

- Hedging S_t exposure with futures (basis risk)
- Types of basis risk
 - Product risk: futures underlying may not match asset
 - Examples
 - Crude oil futures as a hedge for refined products
 - ES futures as hedge for equity market risk
 - Timing risk
 - futures expirations may not match timing of risk exposure
 - Optionality in the timing if delivery is taken
 - Location risk: physical delivery may require additional transport

Poll

Daily Change

Futures vs. Forwards

- X_0 forwards bought at price F_0 at time 0 expiring at T .
- At time t , reposition to X_t , so change position by $X_t - X_0$.
 - Original position X_0 at time t :

$$V_t = X_0(F_t - F_0)P(t, T)$$

Where $P(t, T)$ is a discount factor to shift the time T value back to t

- Costs nothing to enter new position.
- Terminal value of long from time 0, short from t : $(S_T - F_0) - (S_T - F_t)$
- Futures entered at price F_0 has its value updated to 0 every day with a corresponding cash flow. $V_t = 0$.